# TIME SERIES ANLYSES

## Advanced Drug Utilization Research - ISPE 2026

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# Slides & script

https://github.com/martinkhho/ispe-time-series-2026

# What you will learn

By the end of this session, you should be able to:

- Explain what is a **time series**
- Explain how **time series analyses** can be used in drug utilization research
- Use **regression** and **ARIMA** to model time series
- Conduct **interventional ARIMA** analyses

# Disclosures

- I have no personal or financial relationships relevant to this presentation
- Some slides were adapted from last year's talk by Dr. Mina Tadrous

## Declaration of AI-assisted technologies

All instructional material in this presentation was written by the author except where external sources are cited. OpenAI Codex (GPT-5.6-Sol, reasoning: ultra) helped format the slide deck. The author reviewed, edited, and verified the formatting suggestions as needed and takes full responsibility for the final product.

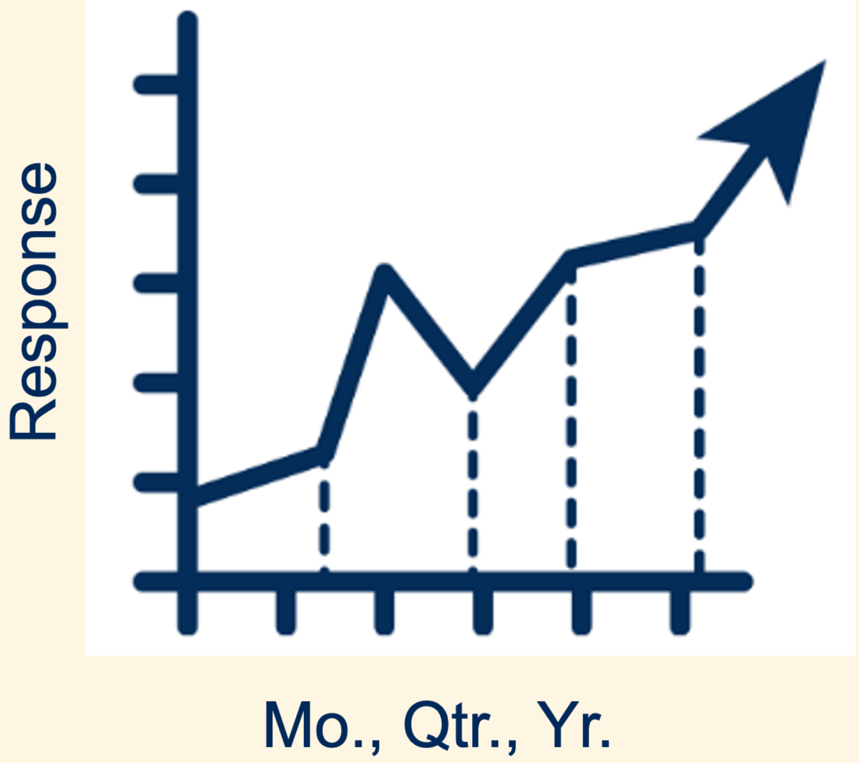
# Univariate time series

- A set of observations of the same variable, indexed by time
- Usually, at regular time intervals
- **Key difference from usual data:** observations are not necessarily independent

$$
\begin{aligned}
x_t &= x_{t-1} + w_t \\
&= (x_{t-2} + w_{t-1}) + w_t
\end{aligned}
$$

# Time series analyses

- Used across many fields, most commonly in finance and economics
- From pharmacopeidemiology lens: a repeated cross-sectional analysis



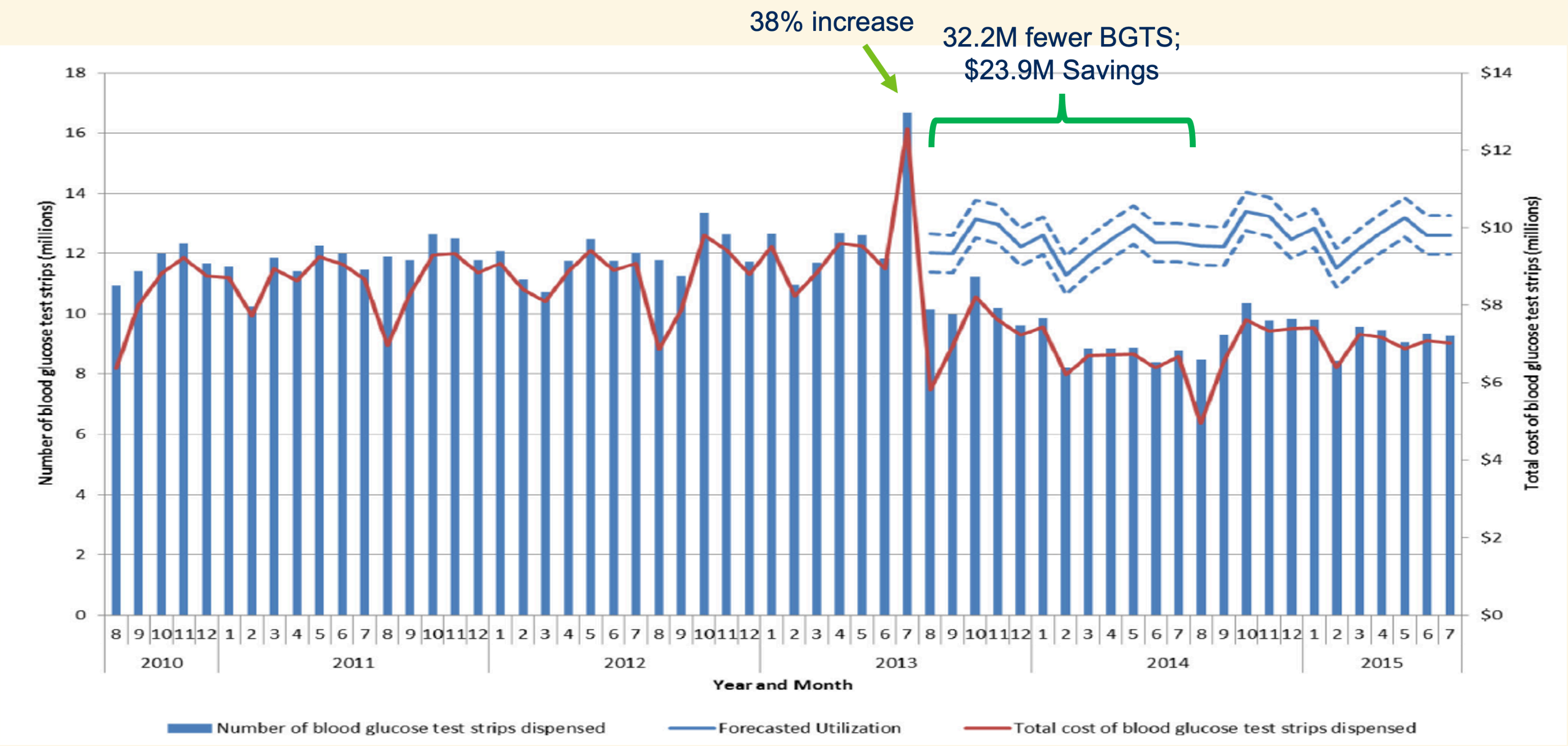
# Motivating example[1] - background

- The province of Ontario, Canada, introduced limits on blood glucose test strips (BGTS) based on diabetes therapy in 2013
- Concern over the impact of this policy on:
  - Test strip utilization and spending
  - Clinical outcomes

# Motivating example - 2013 policy

| Diabetes group | Test strip limits |
|---|---|
| Insulin users | Max 3,000/year |
| Hypoglycemia-inducing oral antihyperglycemics | Max 400/year (~30/month) |
| All others | Max 200/year (~16/month) |

# Motivating example - BGTS utilization & costs




BGTS utilization and costs in Ontario

# Motivating example - clinical outcomes

## Design

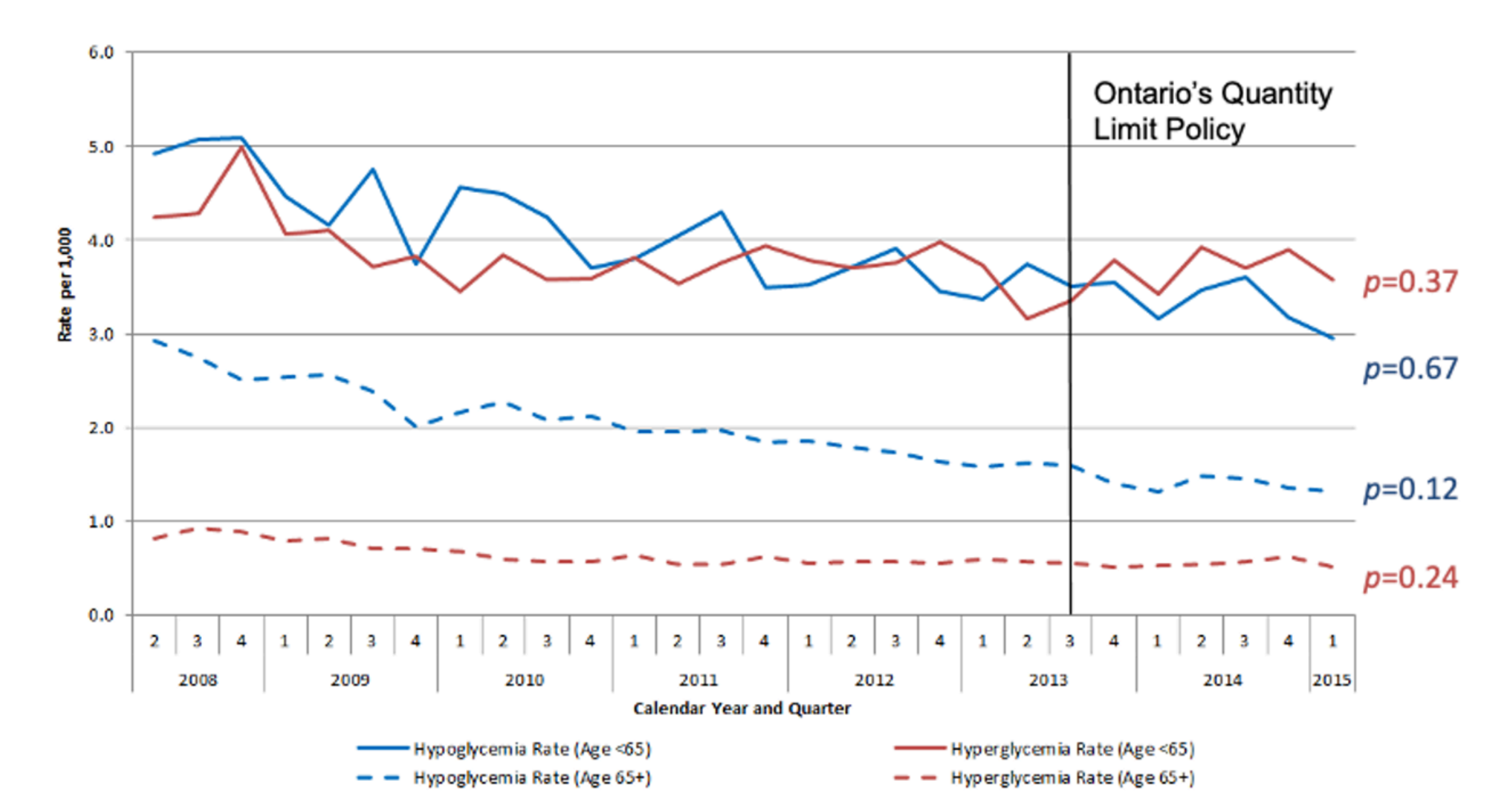
- Repeated cross-sectional analysis
- April 1, 2008 - March 31, 2015 (quarterly)
- Using interventional ARIMA models
- Stratified by age: <65 vs. 65+ years

## Definitions

Users were hierarchically defined based on their diabetes drug therapy in the prior year to the end of the quarter:

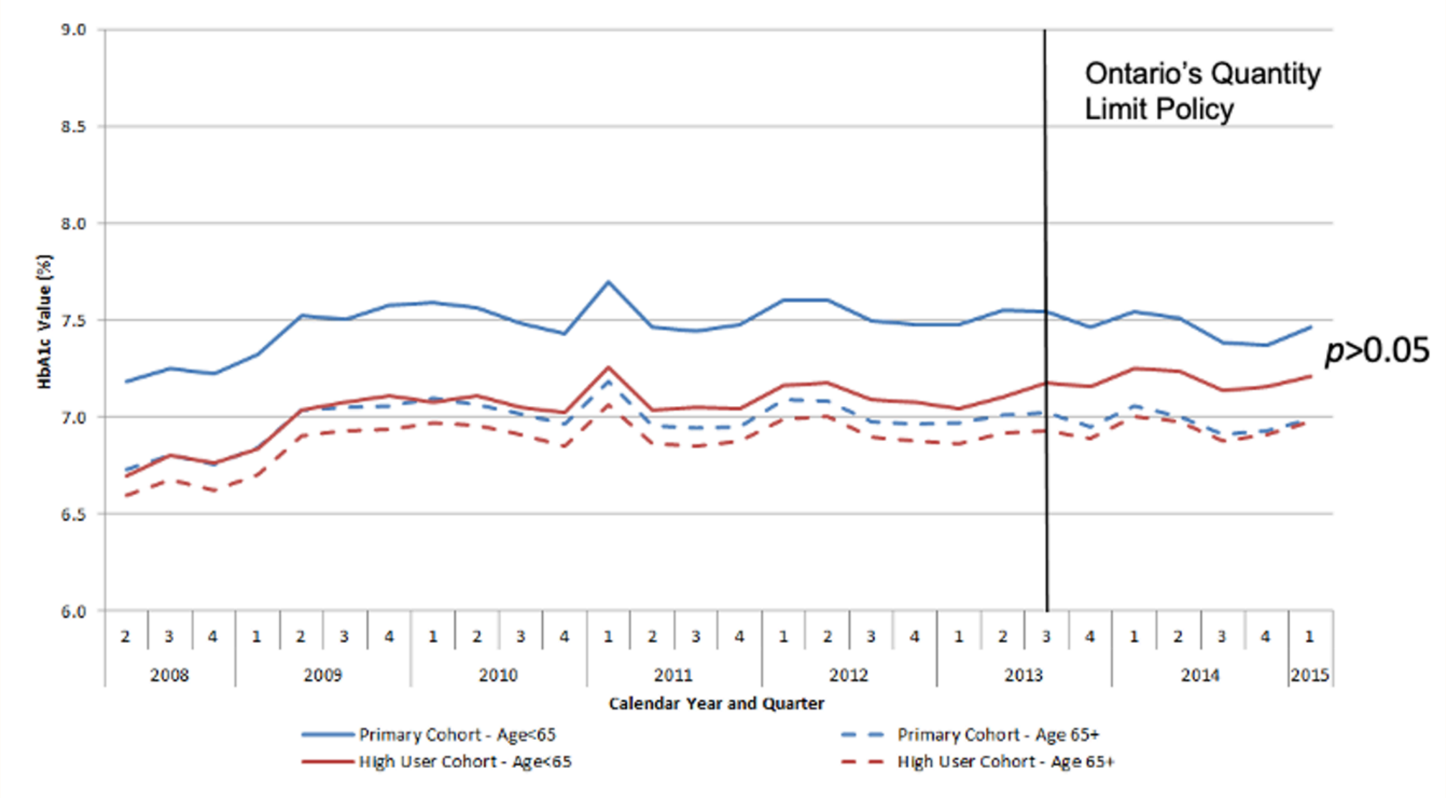
1. Insulin
2. Hypoglycemia-inducing oral antihyperglycemic
3. Other oral antihyperglycemic
4. No drug therapy

# Motivating example - emergency department visits




Rates of emergency department visits for hypoglycemia and hyperglycemia

# Motivating example - mean HbA1c




Mean HbA1c (overall and among high-use cohort)

# Time series characteristics

Trend

Seasonality

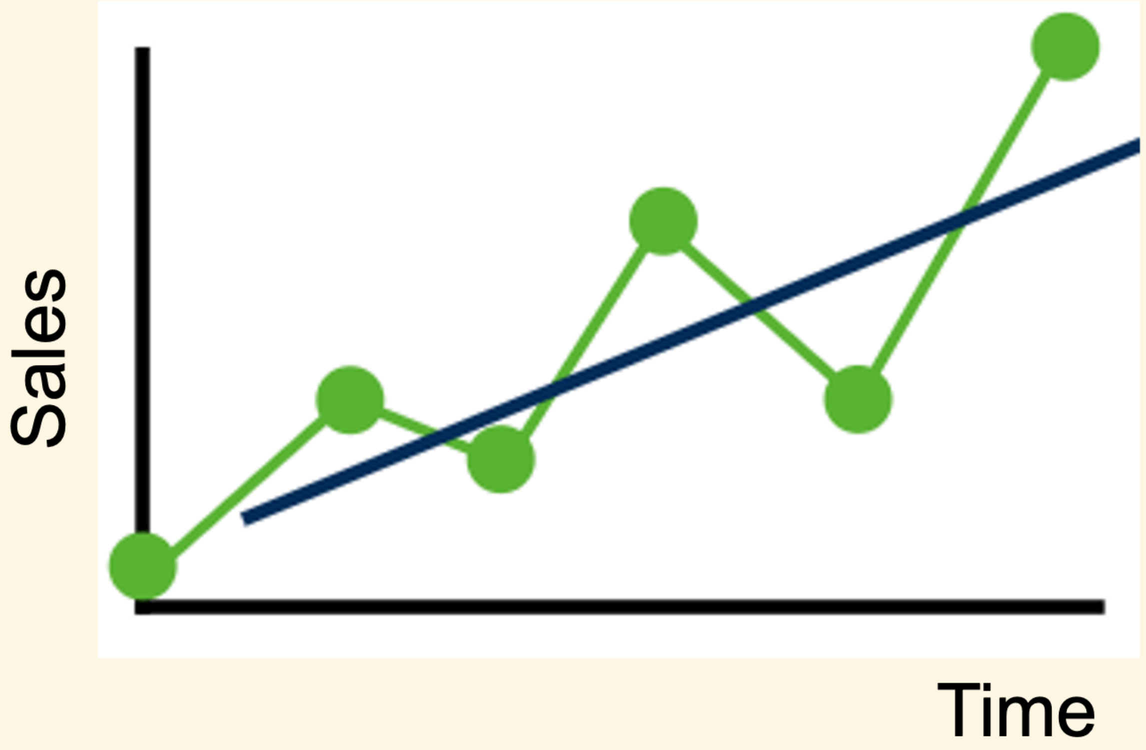
Time

Cycle

Autocorrelation

# Trend

- A persistent, overall upward or downward pattern
- May be due to population growth, shifting prescribing patterns, etc.



# Seasonality

- Regular up and down fluctuations
- May be due to weather, prescription refills, etc.
- Occurs within one year (unlikely to have seasonality if you have annual data)

# Cycles

- Upward and downward swings
- E.g., housing sales, immigration patterns
- Unlike seasonality, may vary in length, often lasting years on end

# Autocorrelation

- The correlation between a data point and a lagged version of itself
- Can be positive or negative
- E.g., stock market, “hot/cold streaks” in sports

# Other

## Interventions

- Policy/formulary change
- Supply chain disruption
- Pandemic

## Irregularities/randomness

- Not systematic
- Residual fluctuations
- Unrepeating

# Time series analyses

## What kinds of questions do we ask?

- Assess impact of interventions
- Forecasting
- Descriptive analysis
- Trend analysis

# Time series analyses

## Approaches

- Regression models that regress $x_t$ on $t$
- ARIMA models that relate the present value to a series of past values and shocks

In ARIMA, the idea is to iteratively model the time series process - you'll know when you're successful because the residuals will be patternless **white noise**:

- Mean 0
- Variance constant
- Uncorrelated

# Regression

$$x_t = f(t)$$

- **Trend:** e.g., linear $t$, quadratic $t^2$
- **Seasonality:** an indicator variable for each season

For example, a time series with **linear trend** and **quarterly seasonality** may be modelled with:

$$x_t = \beta_1 t + \alpha_1 S_1 + \alpha_2 S_2 + \alpha_3 S_3 + \alpha_4 S_4 + \epsilon_t$$

# ARIMA

- **AR:** autoregression
- **I:** integrated
- **MA:** moving average

# Autoregression (AR model)

AR(1) means that we use the last 1 observation to predict the present time:

$$
\begin{aligned}
x_t &= \delta + \phi x_{t-1} + w_t \\
&= \delta + \phi(\delta + \phi x_{t-2} + w_{t-1}) + w_t \\
&\dots
\end{aligned}
$$

- $\delta$ is the intercept
- $\phi$ is the slope (persistence); $|\phi| < 1$ for stationary series, meaning the influence of previous lags decays over time
- $w_t \overset{iid}{\sim} N(0, \sigma_w^2)$

# Autoregression (AR model)

| $t$ | $x_t$ | $x_{t-1}$ |
|---|---|---|
| 1 | 8 | * |
| 2 | 10 | 8 |
| 3 | 11 | 10 |
| 4 | 7 | 11 |
| 5 | 9 | 7 |

- The present value carries information from everything before, because the relationship is transmitted through intermediate lags

# Autoregression (AR model)

- AR(1)

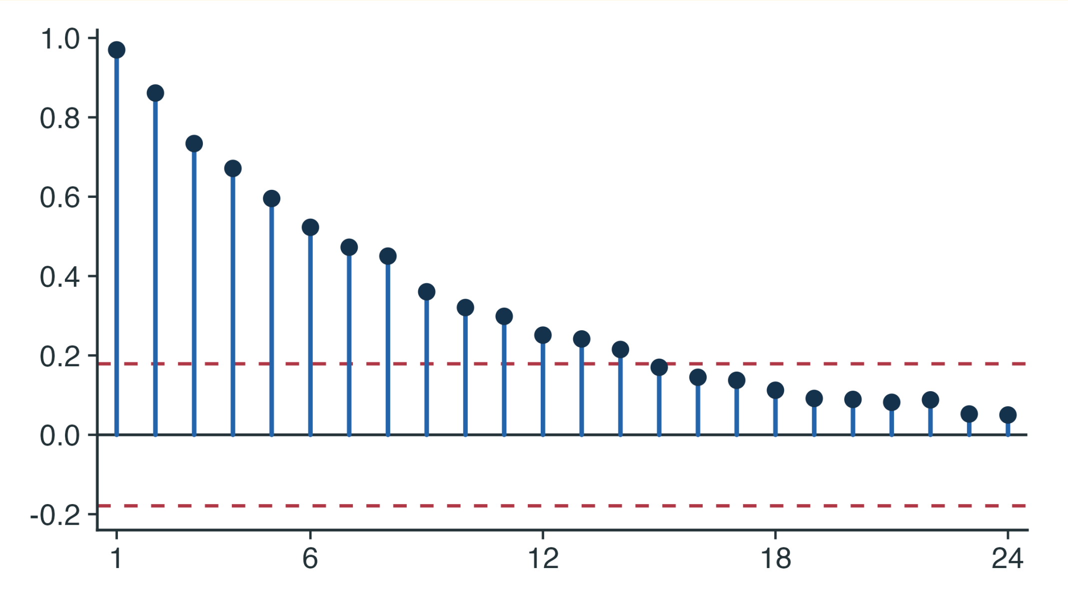
$$x_t = \delta + \phi_1 x_{t-1} + w_t$$

- AR(2)

$$x_t = \delta + \phi_1 x_{t-1} + \phi_2 x_{t-2} + w_t$$

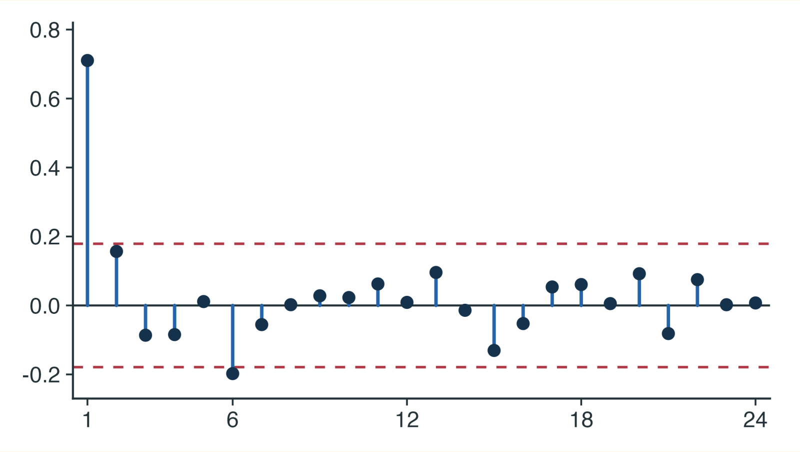
# Autocorrelation function (ACF)

- Correlations between $x_t$ and lagged values $(t-1, t-2, t-3, \ldots)$
- Useful for applying to both the raw series, as well as the model residuals



# Partial autocorrelation function (PACF)

- Correlation between $x_t$ and $x_{t-h}$ after conditioning on the values in between
- Does $x_{t-h}$ have a direct effect on $x_t$ that is not via intermediate values?




- Very useful for determining AR orders
- PACF will spike at the corresponding lag, then shut off to zero afterwards

# Stationarity

- A series is (weakly) stationary if:
  - Mean constant
  - Variance constant
  - Covariance between $x_t$ and $x_{t-h}$ is the same for all $t$ at each lag $h$
- White noise is a special type of stationarity
- Stationarity may be addressed with **differencing**

ACF and PACF patterns (later) will not have their standard interpretations until the series is stationary. ACF of a non-stationary series will decay very slowly.

# Stationarity

## Trend violates stationarity

Augmented Dickey-Fuller (ADF) test:

$p > 0.05$ suggests a need to **difference**.

## Seasonality violates stationarity

Osborn-Chui-Smith-Birchenhall (OCSB) test:

$p > 0.05$ suggests a need to **seasonally difference**.

# Differencing

- Think of it as differentiation but for a discrete variable
- First differencing:

$$\nabla x_t = x_t - x_{t-1}$$

- Second differencing:

$$\nabla^2 x_t = \nabla x_t - \nabla x_{t-1}$$

- We often need first differencing, and rarely more than second differencing
- Over-differencing can introduce autocorrelation where none existed before

# Seasonal differencing

- Seasonal (here, monthly) differencing:

$$\nabla^{12} x_t = x_t - x_{t-12}$$

- Seasonality skips intermediate lags
- Not the same as twelfth differencing

# Moving average (MA model)

- AR processes are not the only way to generate autocorrelation
- While AR captures autocorrelation in past *values* (e.g., $x_{t-1}$ ), MA captures autocorrelation in past *shocks* (e.g., $w_{t-1}$ )
- Shocks are prediction errors

- In AR models, the present value carries information from everything before
- In MA models, the present value carries information from a set number of lags

# Moving average (MA model)

- MA(1)

$$x_t = \mu + \theta_1 w_{t-1} + w_t$$

- MA(2)

$$x_t = \mu + \theta_1 w_{t-1} + \theta_2 w_{t-2} + w_t$$

# ARIMA

- Can include AR terms, differencing operations, and MA terms
- Technically, “IARMA” in the sense that you should difference (if needed) first, to ensure stationarity, before figuring out AR and MA terms
- Without seasonality: $ARIMA\ (p, d, q)$
- With seasonality: $ARIMA\ (p, d, q) \times (P, D, Q)_S$

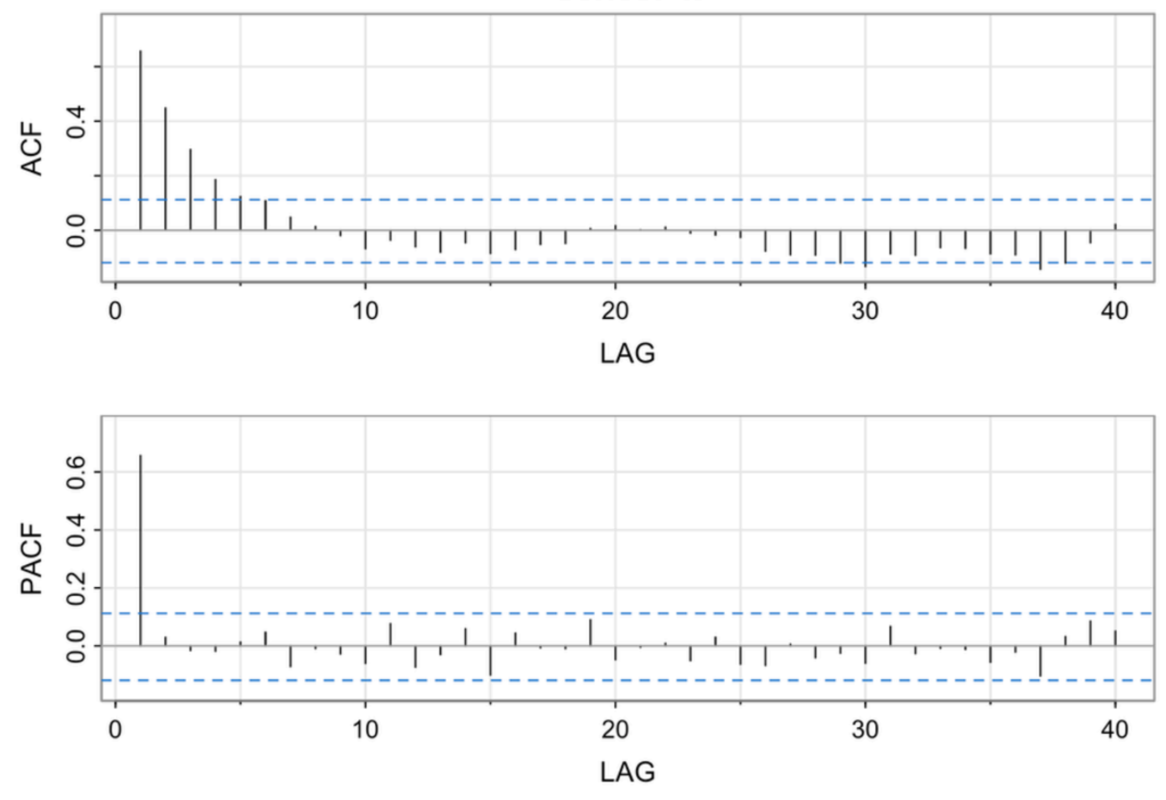
# Correlogram patterns

| True model | ACF | PACF |
|---|---|---|
| AR | Tapers | Cuts off past $p$ |
| MA | Cuts off past $q$ | Tapers |
| ARMA | Tapers | Tapers |

- May not see clean patterns in sample data
- Sign (+/-) of autocorrelation does not matter for identifying orders
- If all lags are non-significant, suggests that series is white noise

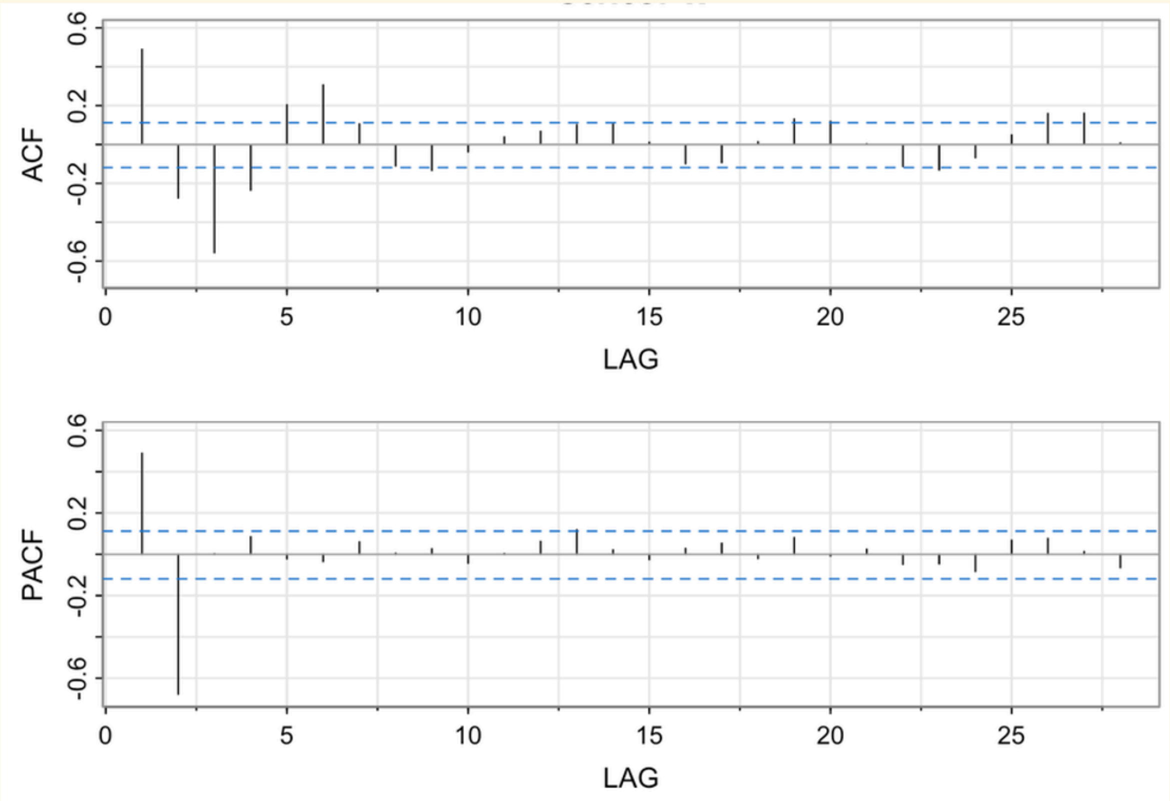
# Identify the model:[2] #1

| True model | ACF | PACF |
|---|---|---|
| AR | Tapers | Cuts off past $p$ |
| MA | Cuts off past $q$ | Tapers |



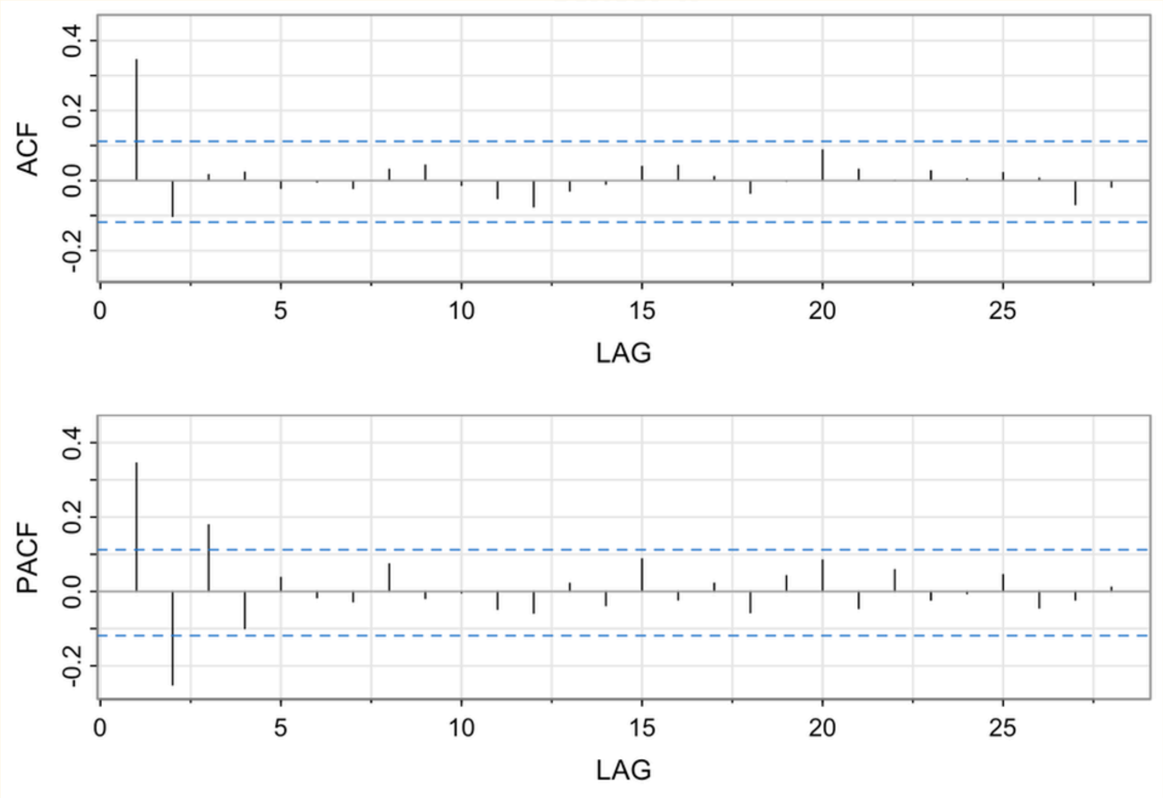
# Identify the model:[2] #2

| True model | ACF | PACF |
|---|---|---|
| AR | Tapers | Cuts off past $p$ |
| MA | Cuts off past $q$ | Tapers |



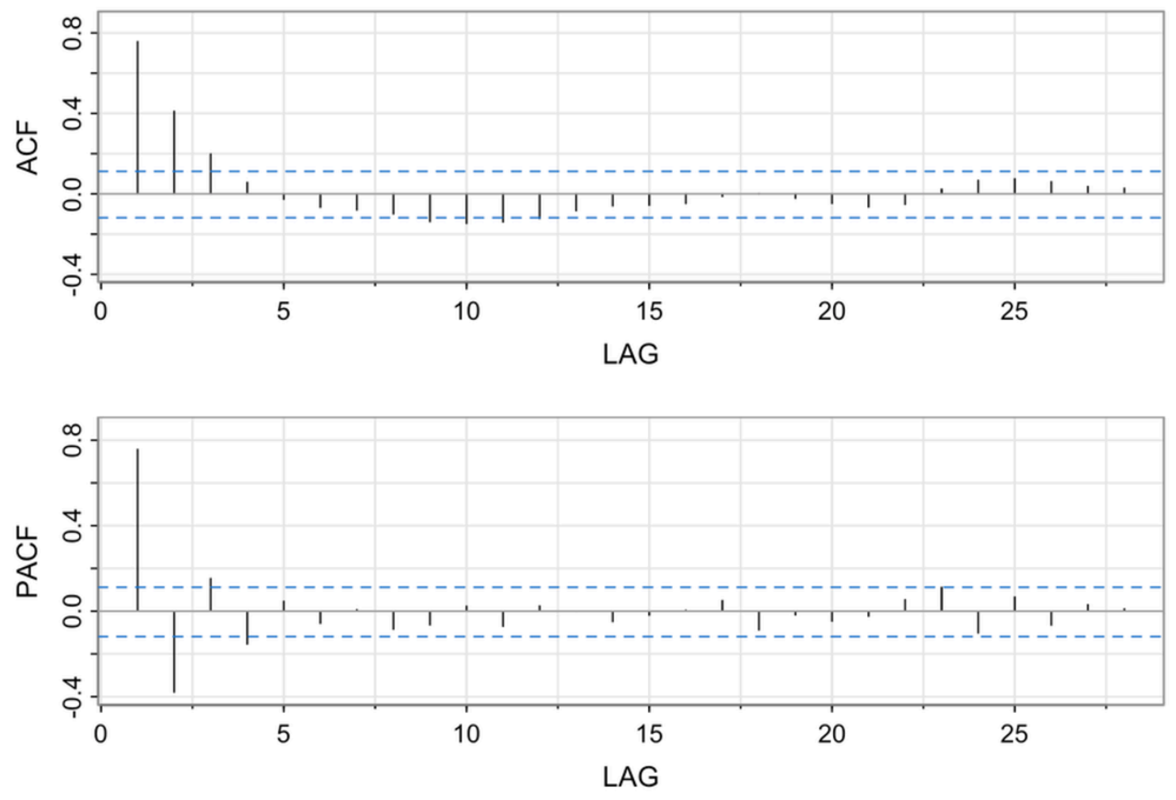
# Identify the model:[2] #3

| True model | ACF | PACF |
|---|---|---|
| AR | Tapers | Cuts off past $p$ |
| MA | Cuts off past $q$ | Tapers |



# Identify the model:[2] #4

| True model | ACF | PACF |
|---|---|---|
| AR | Tapers | Cuts off past $p$ |
| MA | Cuts off past $q$ | Tapers |



# Diagnostics

| Check: | What do we want to see? |
|---|---|
| Coefficients | Significant |
| ACF plot of the residuals | Not significant |
| Ljung-Box test for white noise of the residuals | Not significant |
| Q-Q plot of the residuals | Normality |
| Histogram of the residuals | Normality |
| Residuals-versus-fits plot | No heteroskedasticity |
| Time series plot of the residuals | No heteroskedasticity |
| Forecasts | Reasonable (e.g., not decreasing when expect to increase) |

- Try a different ARIMA model if unsatisfactory

# Comparing models

- Parsimony (low orders)
- Smallest standard errors of forecast values
- Best model fit, while not being overly complex - smaller values of:
  - Akaike information criterion (AIC)
  - AIC with correction for small sample size (AICc)
  - Bayesian information criterion (BIC)

# ARIMA steps

1. Plot the time series

- Pay attention to seasonality and trends, informed also by OCSB and ADF tests
- If both seem possible, do seasonal differencing first as seasonality tends to dominate, then re-evaluate
- If heteroskedasticity, consider log transform

# ARIMA steps

2. Plot correlograms

- Examine the ACF and PACF
- Glean AR and MA terms at low lags, and seasonal AR and MA terms at lags at multiples of $S$

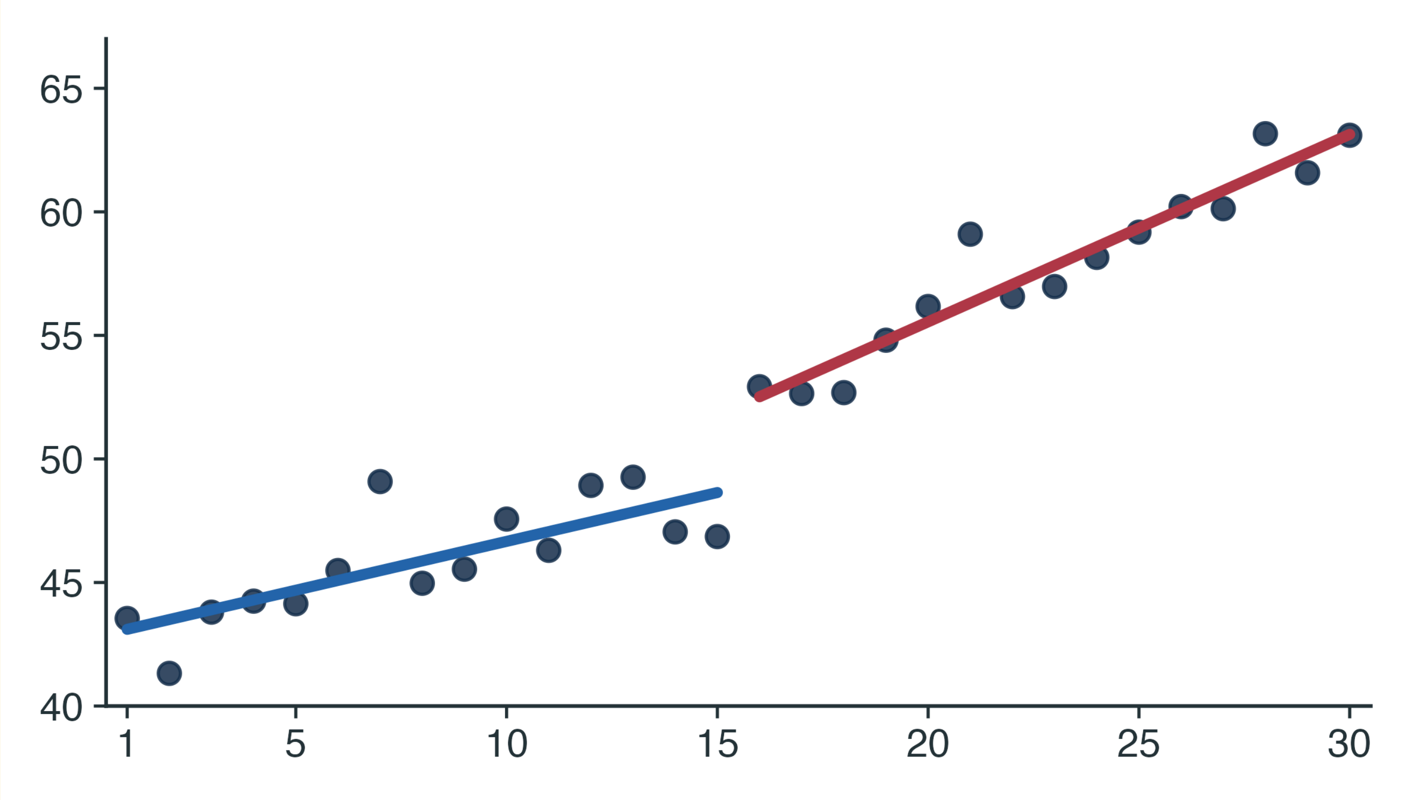
3. Estimate the model
4. Perform diagnostics and select the best model • Iterate if necessary

# Intervention analysis

- Goes beyond modelling the time series; also estimate the effect of an intervention (e.g., policy, formulary change, supply chain disruption, pandemic)
- Ecological; in Ontario, Canada, we use administrative claims data stored at ICES
- **Quasi-experimental studies (QES)** (intervention is as good as random, because assignment is plausibly unrelated to the outcome)
- **Interrupted time series (ITS) analysis** is one type of QES; counterfactual is what the time series would have been had the intervention not occurred
  - Segmented regression
  - Interventional ARIMA

# Intervention analysis: segmented regression

$$y_t = \beta_0 + \beta_1 t + \beta_2 I_t + \beta_3 (t \times I_t) + \epsilon_t$$



# Intervention analysis: segmented regression

Let's break this down into two regression lines (set $t = 0$ for time of intervention):

$$\text{Before intervention } (I_t = 0): y_t = \beta_0 + \beta_1 t + \epsilon_t$$
$$\text{After intervention } (I_t = 1): y_t = (\beta_0 + \beta_2) + (\beta_1 + \beta_3)t$$

- $\beta_0$ is the pre-intervention level (extrapolated) at the time of intervention
- $\beta_1$ is the pre-intervention slope (trend)
- $\beta_2$ is the immediate change in level at the intervention
- $\beta_3$ is the change in slope (trend) following the intervention
- $\beta_1 + \beta_3$ is the post-intervention slope

# Intervention analysis: segmented regression

- Can model non-linear deterministic trend with polynomials or splines
- Can model deterministic seasonality using indicator variables

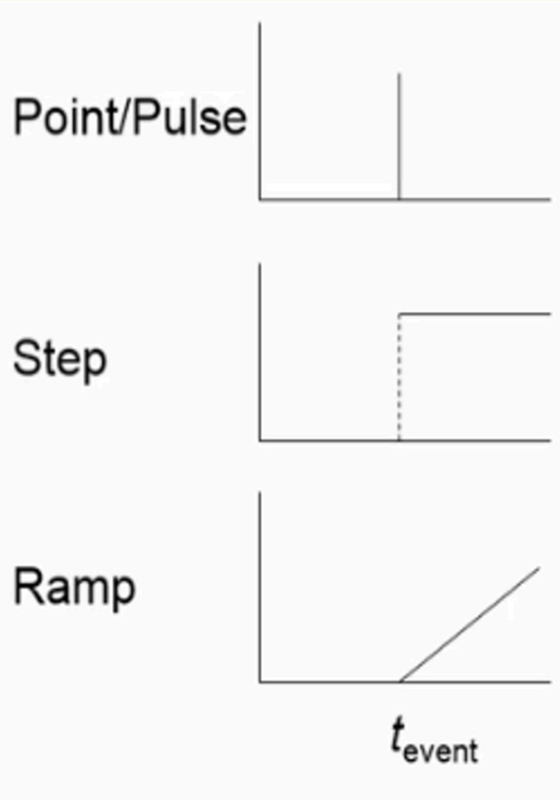
# Intervention analysis: interventional ARIMA

$$y_t = \delta_0 I_t + w_t$$

- $I_t$ is the intervention (see next slide)
- $w_t \sim ARIMA\ (p, d, q) \times (P, D, Q)$

# Intervention analysis: interventional ARIMA[3]

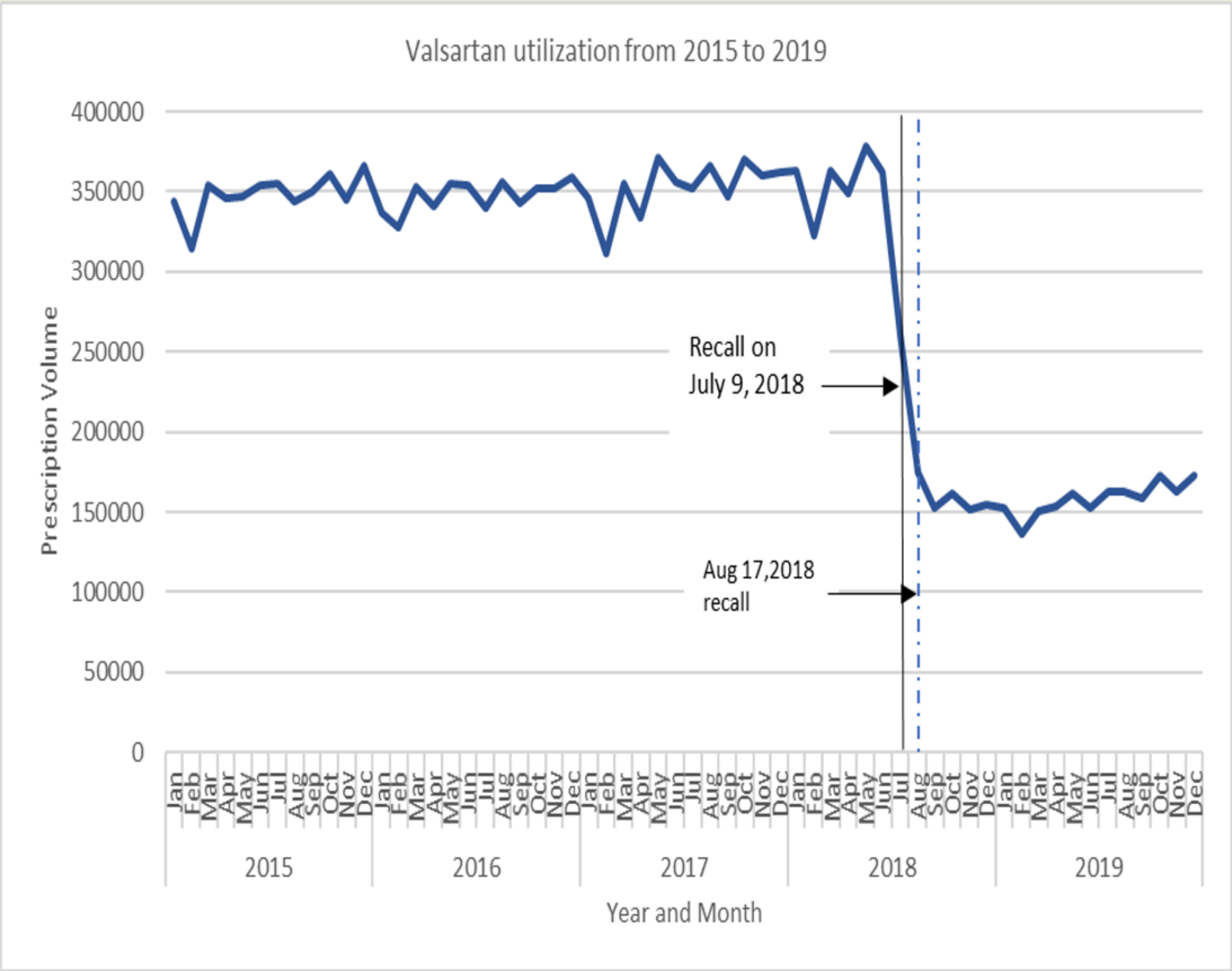
| Intervention | Equation |
|---|---|
| Permanent step | $I_t = \begin{cases} 0, & t < T \\ 1, & t \geq T \end{cases}$ |
| Temporary step over $d$ | $I_t = \begin{cases} 0, & t < T \\ 1, & T \leq t < T + d \\ 0, & t \geq T + d \end{cases}$ |
| Pulse | $I_t = \begin{cases} 0, & t \neq T \\ 1, & t = T \end{cases}$ |
| Ramp | $I_t = \begin{cases} 0, & t < T \\ t - T + 1, & t \geq T \end{cases}$ |




- You can model multiple interventions (e.g., step + ramp)
- Interventions should be pre-specified

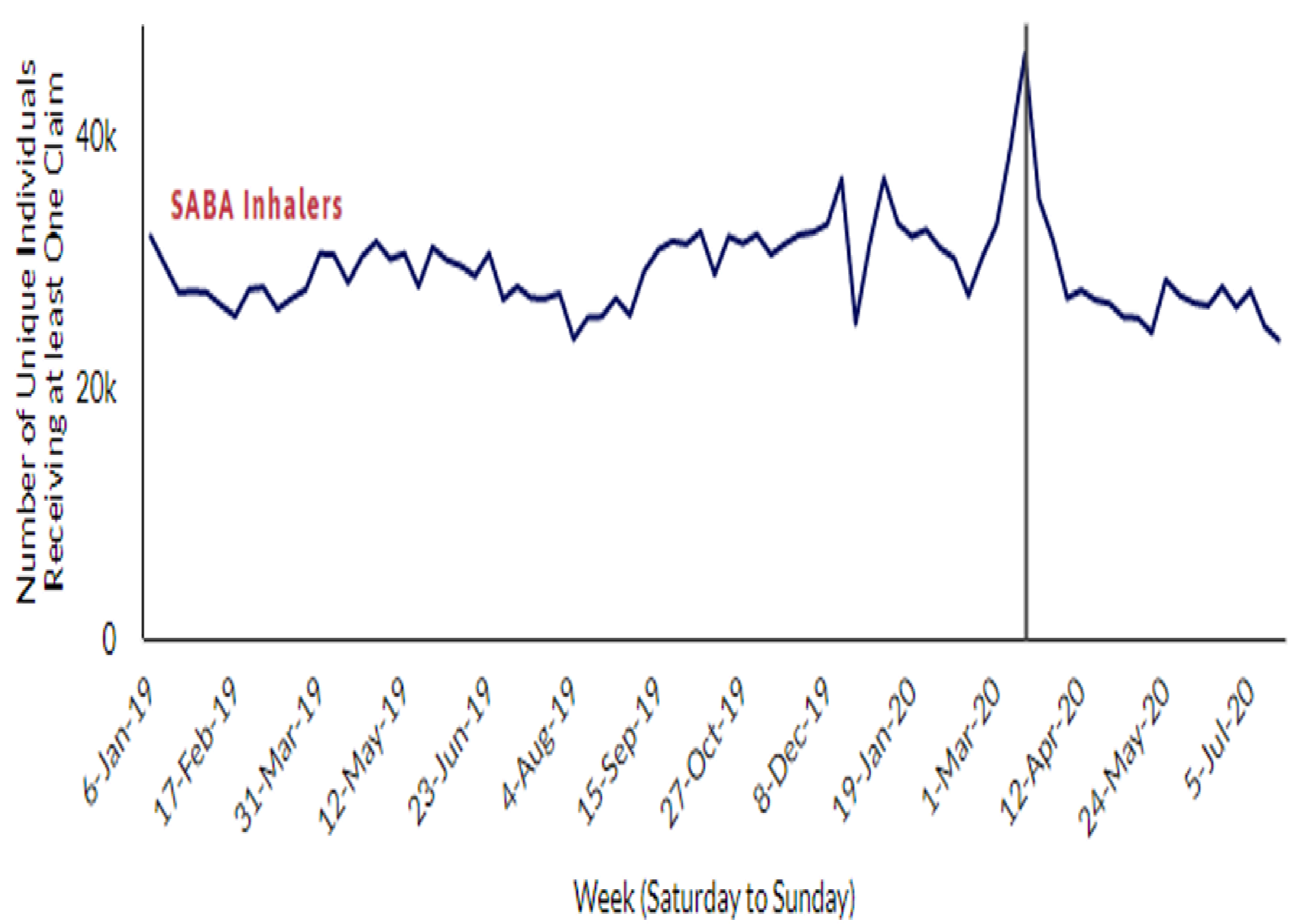
# Name that fit: #1

**Scenario:** Drug recall and shortage in July 2018



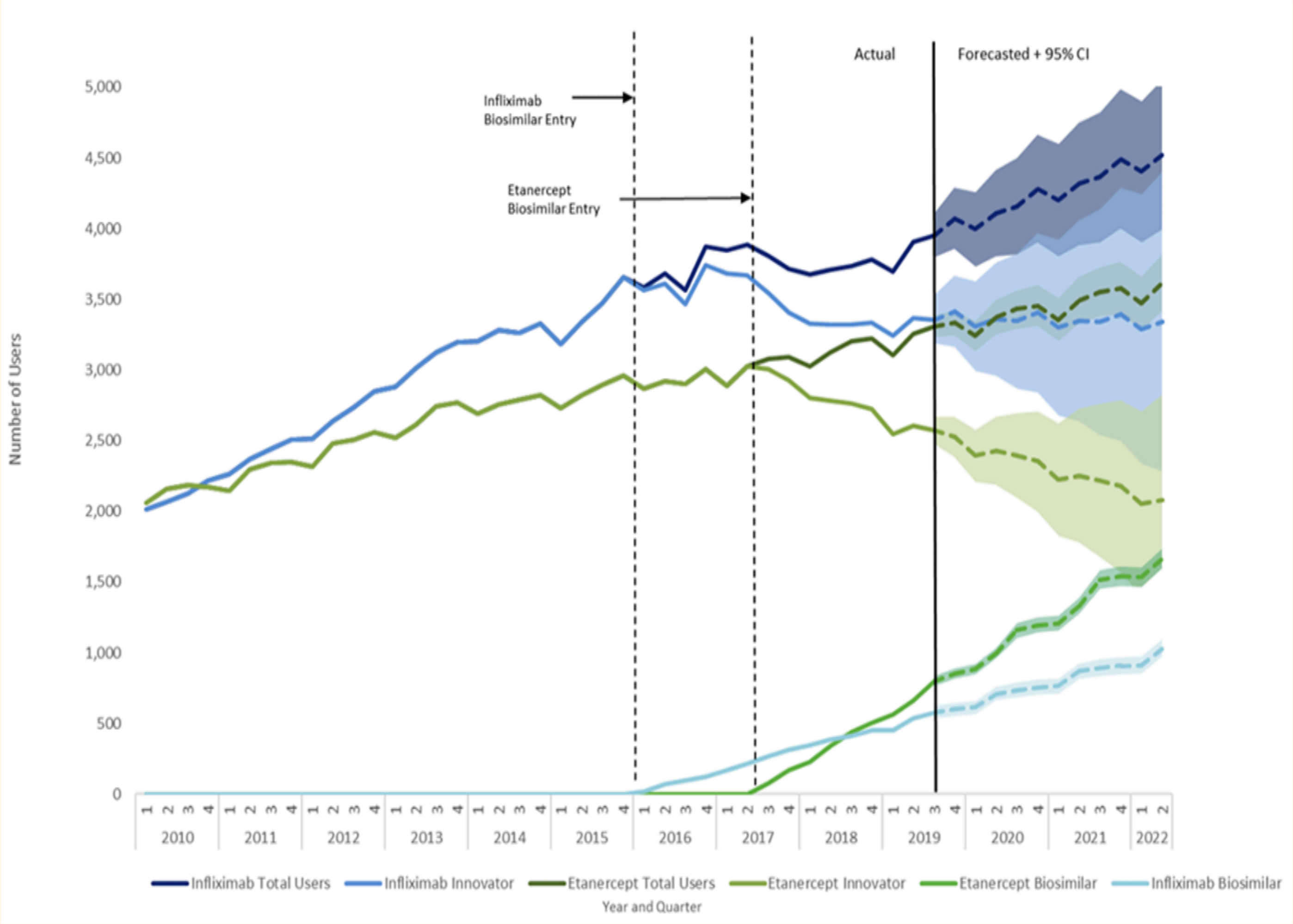
# Name that fit: #2

**Scenario:** COVID-19 pandemic lockdowns impact on salbutamol (albuterol) inhalers



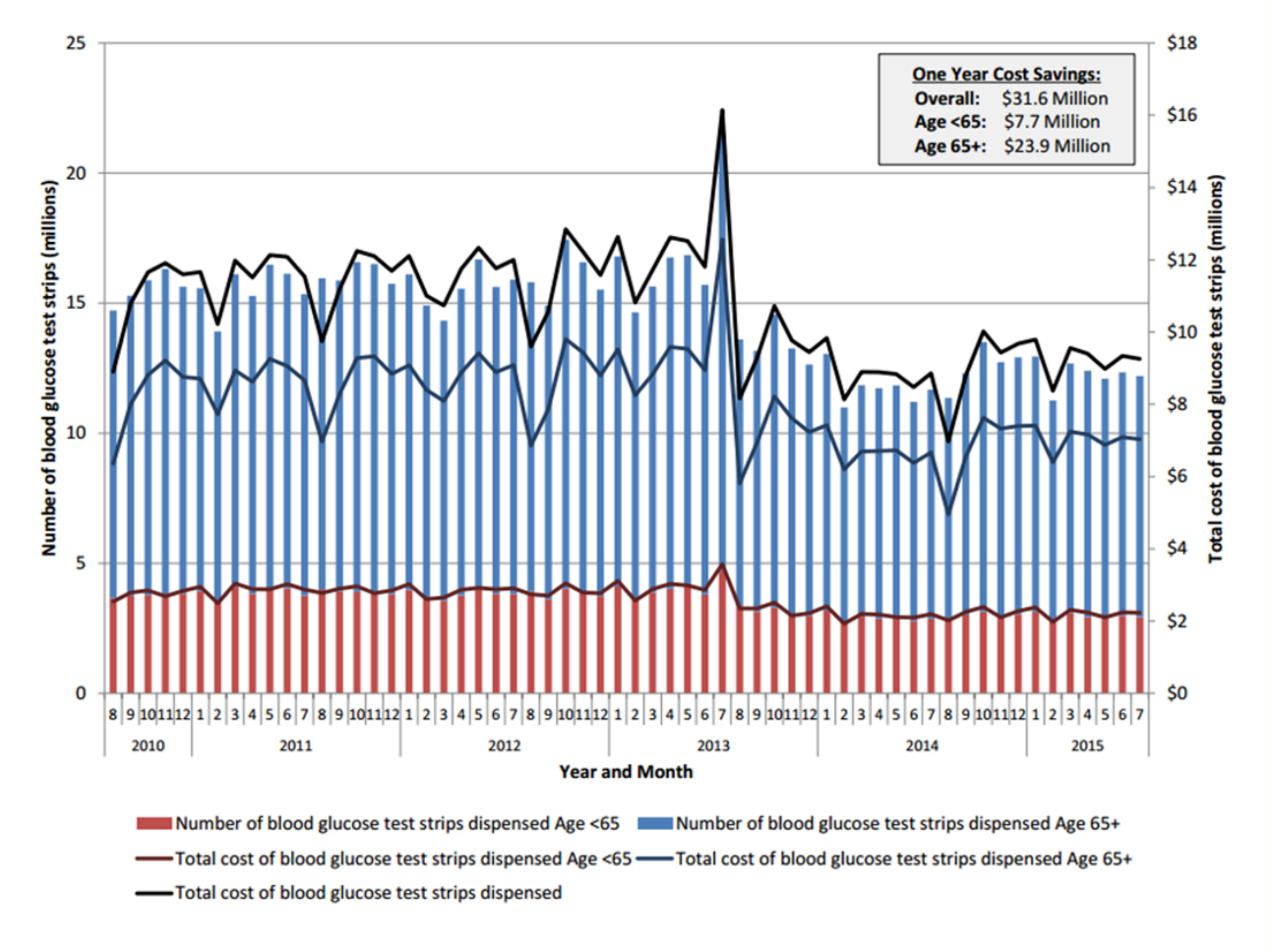
# Name that fit: #3

**Scenario:** Introduction of a new biosimilar on formulary



# Name that fit: #4

**Scenario:** New policy to limit access to a medication class, announced with one month until change



# References


1. Gomes T, Martins D, Tadrous M, et al. Association of a Blood Glucose Test Strip Quantity-Limit Policy With Patient Outcomes: A Population-Based Study. *JAMA Intern Med*. 2017;177(1):61-66. doi:10.1001/jamainternmed.2016.6851

2. PennState Eberly College of Science. STAT 510 Applied Time Series Analysis. Published online 2026. Accessed August 21, 2026. https://online.stat.psu.edu/stat510/Lesson01

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